

UNDERGRUUND

When looking for the next clue, I was considering the implications of the line THE INFORMATION WAS GATHERED AND TRANSMITTED UNDERGRUUND TO AN UNKNOWN LOCATION.

One thing that struck me about this line is the word UNDERGRUUND.

Given everything that I now understand about K1 to K3, this word almost certainly has multiple meanings, in fact the whole line has but one in particular stands out and it is this one I wish to focus upon.

Firstly, we need to rewind a little.

In passage 1, we are shown how rotation works within the cipher. SS AL NI becomes SS NI AL. Here I made a misjudgement because this movement is not only related directly to the movement in passage 2 although it is certainly linked to this.

Within this movement, Mr Sandborn shows us something else that is going on. I will come back to this later.

What this did give us for passage 2 is the characters to look for, and here we find the rotation. AL IN SS. Rotating these 6 characters we get to SS NI AL.

Lets look at the ciphertext K4 for a minute.

```
1  OBKRUXOXGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJQSSEKZZ
2  WATJKLUDIWIWFBNYPVTMTZFPKWGDKZXTJCDIGKUHUAUEKCAR
```

So lets break this into blocks of 6 and rotate each block in the same way were shown. On the left is the cipher and on the right is the cipher flipped.

1 [O, B, K, R, U, O]	[U, O, R, K, O, B]
2 [X, O, G, H, U, L]	[U, L, H, G, X, O]
3 [B, S, O, L, I, F]	[I, F, L, O, B, S]
4 [B, B, W, F, L, R]	[L, R, F, W, B, B]
5 [V, Q, Q, P, R, N]	[R, N, P, Q, V, Q]
6 [G, K, S, S, O, T]	[O, T, S, S, G, K]
7 [W, T, Q, S, J, Q]	[J, Q, S, Q, W, T]
8 [S, S, E, K, Z, Z]	[Z, Z, K, E, S, S]
9 [W, A, T, J, K, L]	[K, L, J, T, W, A]
10 [U, D, I, A, W, I]	[W, I, A, I, U, D]
11 [N, F, B, N, Y, P]	[Y, P, N, B, N, F]
12 [V, T, T, M, Z, F]	[Z, F, M, T, V, T]
13 [P, K, W, G, D, K]	[D, K, G, W, P, K]
14 [Z, X, T, J, C, D]	[C, D, J, T, Z, X]
15 [I, G, K, U, H, U]	[H, U, U, K, I, G]
16 [A, U, E, K, C, A]	[C, A, K, E, A, U]
17 [R]	[R]

At first glance, nothing too exciting but we do have a lot more Us.

Next advance the cipher by one character.

```
1 [O, R, K, O, B, U]
2 [L, H, G, X, O, I]
3 [F, L, O, B, S, L]
4 [R, F, W, B, B, R]
5 [N, P, Q, V, Q, O]
6 [T, S, S, G, K, J]
7 [Q, S, Q, W, T, Z]
8 [Z, K, E, S, S, K]
9 [L, J, T, W, A, W]
10 [I, A, I, U, D, Y]
11 [P, N, B, N, F, Z]
12 [F, M, T, V, T, D]
13 [K, G, W, P, K, C]
14 [D, J, T, Z, X, H]
15 [U, U, K, I, G, C]
16 [A, K, E, A, U, R]
17 [U]
```

If we look down each column, we then find the characters [I, G, T, U, U, L]. These are at respective rows 10, 13, 9, 10, 16, 3

If you're looking at this in rendered form, add 1 to each row - I wrote this based on python output, not markdown line numbers.

INFORMATION GATHERED TRANSMITTED UNDERGRUUND UNKNOWN LOCATION

Lets move those to the top.

```
1 [I, G, T, U, U, L]
2 [P, J, I, N, B, R]
3 [F, U, B, V, O, O]
4 [K, K, T, P, S, J]
5 [D, R, W, Z, B, Z]
6 [U, H, T, I, Q, K]
7 [A, L, K, A, K, W]
8 [U, F, E, O, T, Y]
9 [O, P, K, X, S, Z]
10 [L, S, G, B, A, D]
11 [F, S, O, B, D, C]
12 [R, K, W, V, F, H]
13 [N, J, Q, G, T, C]
14 [T, A, S, W, K, R]
15 [Q, N, Q, S, X, U]
16 [Z, M, E, W, G, I]
17 [L]
```

There are many ways we can reach the next part and Im pretty sure only one will

be accurate. The one Im going to show though, contains both GRUUND (reversed), IGTUUL and the initials JS although there are certainly other ways to make at least 2 of these (I've not yet found another with all 3 together).

First we need to flatten the previous result, then flip every block of 6 the other way, then rotate this by 7 characters.

```

1 IGTUULPJKNBRFUEVOOKKKPSJDRGZBUHOIQKALWAKWUFQOTY
2 OPSXSZLSQBADFSEBDCRKTVMFHNJIGTCTABWKRQNTSXUZWML
3
4 ULUTIGBRNKPJOOVEFUSJPKKKBZZGDRQKIOUHKWAWALTYOQUF
5 SZXSOPADBQLSDCBEFSEFHVTRKTCGINJKRWBTAUSTQNGIWWZML
6
7 RNKPJOOVEFUSJPKKKBZZGDRQKIOUHKWAWALTYOQUF SZXSOPA
8 DBQLSDCBEFSEFHVTRKTCGINJKRWBTAUSTQNGIWWZMLULUTIGB

```

finally rotate the columns by 8, 0, 15, 4, 3, 13

```

1 [D, N, U, U, R, G]
2 [C, V, K, A, H, L]
3 [H, P, E, U, L, G]
4 [C, Z, K, O, F, O]
5 [R, I, G, L, P, S]
6 [U, A, O, F, S, B]
7 [I, O, W, R, S, Q]
8 [U, X, Q, N, K, K]
9 [B, B, S, T, J, T]
10 [R, B, Q, Q, A, S]
11 [O, V, E, Z, N, A]
12 [J, G, T, T, M, D]
13 [Z, W, I, P, I, F]
14 [K, S, B, F, J, T]
15 [W, W, T, K, U, K]
16 [Y, L, W, D, K, X]
17 [Z]

```

GRUUND is row 0 reverse, INFORMATION GATHERED TRANSMITTED UNDERGROUND UNKNOWN LOCATION is at 7 — 12 — 12 — 3 — 15 — 3 and Jim Sandborns initials are diagonal across rows 9 and 10 in columns 5 and 6.

Maybe theres a purer version. Ive not found it yet but Im still looking.

The other rotation

Next: Regarding the other rotation.

There has been a lot of chatter about the word KRYPTOS in the K4 ciphertext and some have posited that there might be a rotation going on.

Well thats exactly what there is. SS AL NI shows you how this works.

Firstly, its important to note, SS doesnt move whilst AL and NI change locations.

Next, Passage 3 shows you a machine. In this, 4 characters are separated and the rest split into lines 31 characters long.

Lets recap on that:

```
1  SS  THECANDLEANDPEEREDINREDTHELOWER  
2      PARTWITHTREMBLINGHANDSIMADEINTH  
3      EUPPERLEFTHANDCORNERATINYBREACH  AL
```

Now do the same to K4.

```
1  OB  KRUXOGHULBSOLIFBBWFLRVQQPRNGKS  
2      SOTWTQSJQSSEKZZWATJKLUDIAWINFBN  
3      YPVTMZFPKWGDZXTJCDIGKUHUAUEKC  AR
```

Next: Reverse the middle line and rotate it by 3 characters

```
1  OB  KRUXOGHULBSOLIFBBWFLRVQQPRNGKS  
2      TOSNBFNIWAIDULKJTAWZZKESSQJSQW  
3      YPVTMZFPKWGDZXTJCDIGKUHUAUEKC  AR
```

Finally, swap lines 2 and 3.

```
1  OB  KRUXOGHULBSOLIFBBWFLRVQQPRNGKS  
2      YPVTMZFPKWGDZXTJCDIGKUHUAUEKC  
3      TOSNBFNIWAIDULKJTAWZZKESSQJSQW  AR
```

That is KRYPTOS read by rows.